

Accelerated aging tests for evaluations of durability performance of FRP reinforcing bars for concrete structures

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Abstract

This paper presents accelerated aging test results of a durability study on fiber-reinforced polymer (FRP) reinforcing bars for concrete structures. Bare FRP bars and also bars embedded in concrete, primarily for glass composites, were exposed to five different solutions: water, two types of simulated alkaline pore solutions of normal and high performance concrete, saline solution, and combined alkaline solution with chloride ions. The aging was accelerated by using elevated temperatures. Wetting and drying and freezing and thawing cycles were combined with some solutions to simulate the coupling effects as expected in field conditions. The tensile strength and inter-laminar shear strength of FRP bars were determined before and after exposure, and were considered to be measures of durability performance of the specimens. In addition, pullout tests were conducted to investigate the effects of accelerated exposure on the durability of bond strength between FRP bars and concrete. The results showed that when exposed to simulated environments significant strength loss resulted from the accelerated exposure of both bare and embedded GFRP bars, including bond strength, especially for solutions at 60 °C. In contrast carbon fiber-reinforced polymer (CFRP) bars displayed excellent durability performance. For GFRP bars, continuous immersion resulted in greater degradation than exposure to wetting and drying cycling. In contrast, freezing and thawing cycling combined with solutions had little degradation effects on the FRP bars.

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Keywords: Accelerated aging test; Bond strength; Concrete; Durability; FRP; Pullout test

1. Introduction

Corrosion of steel in concrete is a major cause of deterioration of reinforced concrete structures. Fiber-reinforced polymer (FRP) bars are being increasingly used in concrete structures due to their high stiffness to weight ratio, good fatigue properties, ease of handling, and primarily resistance to corrosion [1]. But due to the versatile applications of concrete in different construction fields, FRP reinforced concrete structures may often be exposed to wetting and drying (WD) cycles, sea-salts in hot-humid climates, freezing and thawing (FT) conditions, deicing salts in cold climates, and many other severe environments. In addition

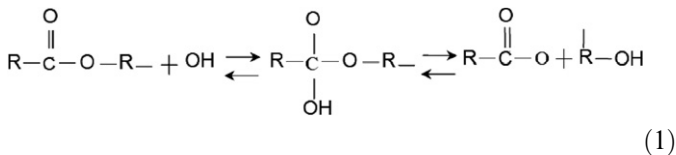
to the above external attack, glass fiber-reinforced polymer (GFRP) bars, which are extensively used in civil engineering structures due to their low cost, are also readily attacked by concrete pore solutions having pH values of 12.4–13.7. Therefore, the evaluations of long-term durability of FRP bars, particularly GFRP, are of primary importance in order to establish the life-cycle performance of FRP reinforced structures.

The degradation in FRP bars starts when free hydroxyl ions (OH^-) and water molecules diffuse through the matrix of FRP bars. Polyester, vinyl ester, and epoxy are the most commonly used resins in FRP bars in civil engineering. The ester group, the weakest bond in polyester and vinyl ester matrices, is prone to degradation by hydrolysis as shown in Eq. (1). The vinyl ester matrix with fewer esters groups is less susceptible to hydrolysis than polyester matrix [2–4]. The epoxy matrix is usually not affected by hydrolysis

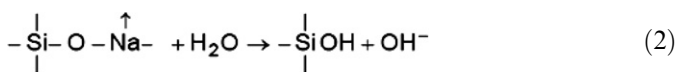
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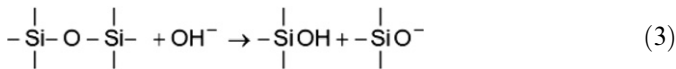
since there is no ester group in its molecular structure. The deterioration of matrices can also occur due to plasticization and swelling.



Glass, aramid and carbon fibers are widely used as fibers in FRP bars. The degradation mechanisms of glass fibers in water or alkaline solution can be classified into two categories: leaching and etching [5]. The diffusion of the alkali ions out of the glass structure, known as “leaching”, is the most important reaction in the dissolution of glass in water as shown in the following equation:



The second important reaction is called “etching”, in which the hydroxyl ions break the Si–O–Si structure as shown in the following equation:



As can be inferred from Eqs. (2) and (3), the dissolution of glass fibers in water would eventually evolve into an alkali attack. All three degradation processes (Eqs. (1)–(3)) are aggravated in presence of alkaline solutions (hydroxyl ions). As a consequence of chemical attack, the resulting formation of hydration products may cause notching and embrittlement of the fibers. As with most organic polymeric fibers, aramid fibers are particularly susceptible to moisture absorption. Carbon fibers are known to be inert to chemical environments and do not absorb water.

The degradation of interphase between fibers and matrix is complicated. The interphase is an inhomogeneous region with a thickness of about one micrometer. It is usually the primary weakest link and can degrade easily. Three damage mechanisms were proposed by Bradshaw and Brinson [6]: (1) matrix osmotic cracking; (2) interfacial debonding; and (3) delamination.

Due to the increased use of FRP bars in concrete structures, the durability performance of FRP bars has been an important research topic in recent years. The effects of various environments such as moisture, saline, and alkaline conditions on the durability performance of FRP bars were investigated by using simulated environments [7–9]. Some studies also investigated the effects of material characteristics such as resin and fiber types, their volume fractions, surface treatment, and manufacturing process on the durability performance of FRP bars [2,3,9,10]. Accelerated test methods with the acceleration factors of elevated temperature ([2,8,9,11,12]) and/or sustained load ([3,8,13]) were often used to save time and cost in durability studies. Based on short term data from accelerated tests, the Arrhenius model was adopted by some researchers ([11,12]) to make

predictions for the long term behavior of the FRP bars, while the Fick’s diffusion law was used to make predictions by other researchers [7,13,14].

2. Research significance

Although a number of durability studies on FRP bars have been reported by various researchers, no general conclusions are possible as researchers used different testing procedures and conditions. In some cases, even conflicting results have been reported [4,15,16]. Limited information [9,17] is available on the exposure of FRP bars in combined WD cycles or FT cycles in aggressive environment, and little information [18] is available on the durability performance of FRP bars exposed to high performance concrete (HPC) environment. Few studies [19,20] have investigated the durability of bond strength between FRP bars and concrete and possibly no attempts have been made to evaluate the bond strength of FRP bars embedded in HPC.

This study presents durability evaluations of different FRP bars under various simulated exposure conditions, with the objective of providing useful guidelines for defining appropriate durability test environments. In addition to bare FRP samples, a few specimens were also embedded in concrete for comparative purposes. Also, bond tests were carried out on FRP bars embedded in both normal concrete (NC) and HPC, leading to significant differences due to concrete type.

3. Experimental program

3.1. Materials

In this study, two types of FRP bars made of E-glass fibers and vinyl ester resin were primarily selected. FRP bars made of carbon fibers and epoxy resin were selected for comparison purpose. The FRP bars were produced by pultrusion and their characteristics are given in Table 1. Both GFRP1 and GFRP2 bars were helically wrapped and slightly sand coated as shown in Fig. 1. The GFRP bars made of E-glass fibers and vinyl ester resin were selected because of their widespread applications in civil infrastructure and potential cost savings. GFRP bars with polyester resin were excluded due to much inferior durability performance [9,20]. For bond strength tests, the FRP bars were embedded in two different types of concrete: NC and HPC. Table 2 shows the mix design and 28-days compressive strength for each type of concrete.

3.2. Simulated environmental conditions

Five different solutions were made to simulate accelerated exposure of specimens to field conditions. Due to the increasing use of new and more durable concrete mixtures, HPC environment was also included. The following solutions were prepared for exposing the FRP bars.

Table 1
Material aspect of FRP specimens

Bar type	Fiber type	Resin type	Bar size	Nominal diameter (mm)	Brand name
GFRP1	E-glass	Vinyl ester	#3 and #4	9.53 and 12.70	Aslan 100
GFRP2	E2-glass	Vinyl ester	#3	9.53	N/A
CFRP	Carbon	Epoxy	#3	9.00	Aslan 200

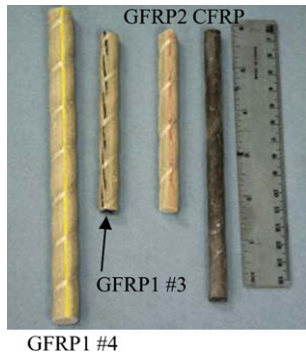


Fig. 1. FRP bars.

Solution 1 was tap water to simulate high humidity and used as a reference environment. Both Solutions 2 and 3 were made with combined sodium hydroxide (NaOH), potassium hydroxide (KOH), and calcium hydroxide ($\text{Ca}(\text{OH})_2$) according to [21,22]. Solution 2 with a pH value of about 13.6 was intended to simulate the pore solution of NC. Solution 3 with a pH value of about 12.7 and different proportions of hydroxides compared to Solution 2, was made to simulate the pore solution of HPC, which is often made by adding supplementary cementitious materials such as fly ash, slag and silica fume. Solution 4 consisted of sodium chloride (NaCl) and sodium sulfate (Na_2SO_4) to simulate ocean water. Solution 5 included sodium chloride (NaCl) and potassium hydroxide (KOH) with a pH value of about 13. This solution with chloride ion (Cl^-)

concentration of 1.23 mol/L was to simulate concrete pore solution contaminated with chloride ions from deicing salts [7]. The compositions of Solution 2 through Solution 5 are listed in Table 3. Elevated temperatures of 40 and 60 °C were used to accelerate the attack of simulated environments on FRP bars, since the degradation rate of FRP bars mainly depends on diffusion rate and chemical reaction rate, both of which can be accelerated by elevated temperatures.

In addition to continuous immersion in the solutions, wetting and drying (WD) cycles were combined with Solution 1 through Solution 4 to simulate the combined effects as expected in real-life conditions. Each WD cycle consisted of four days of immersion at 60 °C followed by four days of drying at 20 °C. The specimens were subjected to a total number of nine WD cycles (72 days). The freezing and thawing (FT) cycles were combined with only Solutions 1 and 5. Each FT cycle consisted of 30 min of soaking at 20 °C, 90 min of ramping from 20 to −20 °C, 30 min of soaking at −20 °C, and finally 90 min of ramping from −20 to 20 °C.

For a realistic evaluation of the durability of FRP bars for concrete structures, a few FRP bars were embedded in NC and HPC. Then one set of each of these concrete-embedded specimens was immersed in Solution 3 at 60 °C and the other specimens were kept in a curing tank (Solution 1) as control.

Eight temperature controlled tanks were custom designed for specimens exposed to Solution 1 through Solution 4 at temperatures of 40 and 60 °C. Each of the

Table 2
Concrete mix design (kg/m^3)

Concrete type	W/Cm	Water	Cement	Sand	Gravel	Slag	Silica fume	AEA (mL)	HWR (mL)	WR (mL)	28-days, compressive strength (MPa)
NC	0.55	175	320	703	1054	–	–	768	–	–	20
HPC	0.35	157.5	292.5	672	999	135	22.5	1570	3100	1200	65

Notes: W/Cm = water to cementitious materials ratio; AEA = air-entraining admixture; HWR = high range water-reducing admixture; WR = water-reducing admixture.

Table 3
Compositions of Solution 2 through Solution 5

Solution no.	Quantities in g/L (mol/L)				
	NaOH	KOH	$\text{Ca}(\text{OH})_2$	NaCl	Na_2SO_4
Solution 2	2.4 (0.06)	19.6 (0.35)	2 (0.027)	–	–
Solution 3	0.6 (0.015)	1.4 (0.025)	0.037 (0.005)	–	–
Solution 4	–	–	–	30	5
Solution 5	–	5.6 (0.1)	–	71.66 (1.23)	–

380 L capacity double walled temperature controlled tanks was made of one piece molded high density polyethylene. Each tank included a Teflon coated immersion heater (500 W/120 V) with riser/junction box and digital temperature control with ± 1 °F accuracy. The control had programmable high set point, sensor short and break protection and security code restricted menu. A typical temperature controlled tank is shown in Fig. 2. For exposure of specimens at room temperature in Solution 1 through Solution 4, polypropylene heavy-wall tanks with diameter of 0.6 m and overall height of 1.2 m were used. Specimens exposed to FT cycles in Solutions 1 and 5 were kept in digital controlled and programmable (CSZ ZH-16) environmental chambers.

3.3. Testing plan

To evaluate the durability performance of FRP bars, unconditioned and conditioned specimens were tested for tensile strength, interlaminar shear strength (ILSS), and bond strength between FRP bars and concrete. The above mechanical properties have been selected on the basis of the following considerations. FRP bars embedded in concrete should meet two basic requirements: (1) the tensile strength must be enough to carry the desired tensile force; (2) the bond strength between FRP bar and concrete must be sufficient to transfer the load. Moreover, the ILSS, which is greatly influenced by the polymer matrix and fiber/matrix interphase, was considered to be closely related to the bond strength between FRP reinforcement and concrete [19]. All these properties are critical for application of FRP bars as reinforcement in concrete structures. The tensile strength test and short beam test were conducted in accordance with ASTM D 3916-94 and ASTM D 4475-85, respectively, with necessary modifications. Since there is no standard test for pullout of FRP bars embedded in concrete, tests were conducted on the basis of methods followed by Bank et al. [19] with necessary modifications.

Tensile test specimens were #3 GFRP1, GFRP2, and CFRP bars (the type and diameter of FRP bars are shown in Table 1) having lengths of 1.02 m. The grip used in this

study was a 200-mm long steel pipe cut lengthwise into two halves. Coated with epoxy in their inner surface, the split pipes were bonded on the ends of each of the FRP bar with hose clamps as shown in Fig. 3. The tests were carried out by gripping the steel pipe into the wedges of a Baldwin machine. The loading time for the test was 2–4 min for GFRP bars and 5–7 min for CFRP bars. Some of the GFRP2 bars for tensile tests were embedded within 100 mm diameter by 200 mm long NC cylinders, within the midlength of the bars, as shown in Fig. 4. The protruded portions of bars were coated with epoxy before the immersion in order to protect them from solutions. Before the tensile tests, FRP bars were extracted from the enclosed concrete cylinders by splitting the concrete.

The specimens for short beam tests were #3 GFRP1, GFRP2, and CFRP bars with span to diameter ratio of 4. The test was carried out in a Material Testing System (MTS) machine. The experimental setup for the short beam



Fig. 3. FRP bars with split steel pipes clamped on both ends for tensile test.

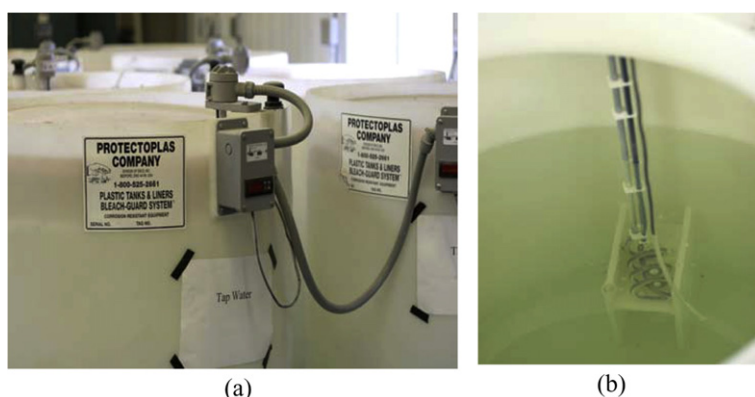


Fig. 2. Temperature controlled tank: (a) digital temperature control; (b) Teflon coated immersion heater.

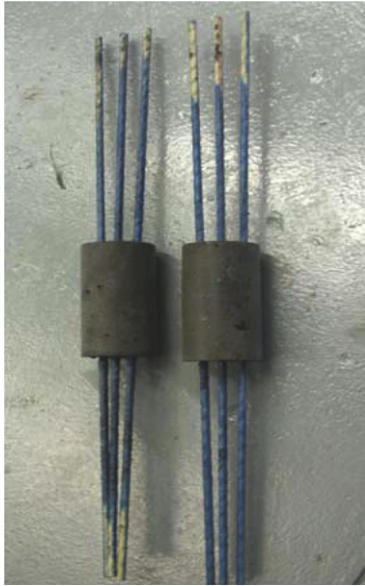


Fig. 4. GFRP2 bars embedded in normal concrete for tensile test.

test is shown in Fig. 5. The loading rate for the test was 1.3 mm/min. The load and movement of the loading head were recorded by the StrainSmart data acquisition system. Some of #3 GFRP1 bars for short beam test were embedded in NC and HPC, respectively. These concrete-embedded specimens were fabricated by aligning the GFRP bars vertically through concrete cylinders using a wooden jig. Then these 152 mm diameter by 305 mm long concrete cylinder specimens (as schematically shown in Fig. 6) were demolded from the wood jig 2 days after casting. After being stored for 28 days in a curing tank, one set of specimens was immersed in Solution 3 at 60 °C for 60 days, and a corresponding set of specimens was kept in the curing tank as controls. After immersion, each of the concrete cylinder specimens was cut into two sections as schematically shown in Fig. 6. The #3 GFRP1 bar extracted from Section 2 was used for short beam test. Section 1 was then used

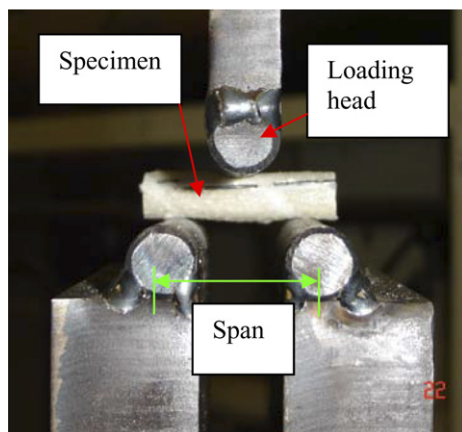


Fig. 5. Experimental setup for short beam test.

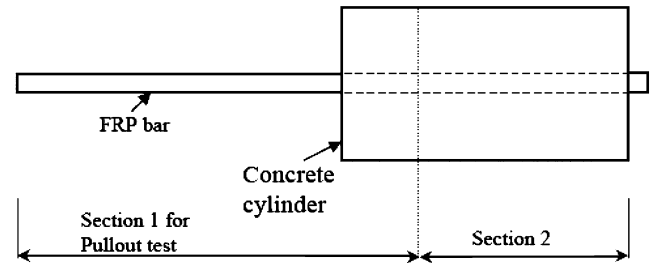


Fig. 6. GFRP1 bars embedded in concrete for pullout and short beam tests.

Table 4
Specimens for pullout test

Bar types	Bar size	Concrete cylinder dimension (diameter × long mm)	Embedded length	Concrete type
GFRP1	#3	152 × 48	5d	NC
			5d	HPC
GFRP1	#4	152 × 64	5d	NC
			5d	HPC

Note: d = bar diameter.

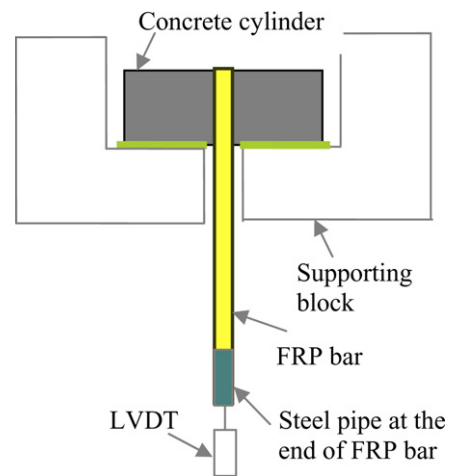


Fig. 7. Experimental setup for pullout test.

as the specimen for pullout test, and the information for these specimens is given in Table 4. The test setup for pullout test is schematically shown in Fig. 7.

The experimental program for evaluation of durability performance of FRP bars in various environments is summarized in Tables 5 and 6.

4. Test results and discussion

Small white blisters were observed on the external surface of GFRP bar after exposure, especially those exposed to solutions at 60 °C as shown in Fig. 8(a). These blisters were also found for GFRP bars embedded in NC as shown

Table 5
Environmental conditions and results for tensile test

Bar types	Environmental exposure			No. of specimens	Mean value of tensile strength (MPa)	COV (%)	Retention
GFRP1 bars	Unconditioned Solution 1	60 °C	70 days	6	771	2	1.00
		40 °C	70 days	6	544	7	0.71
		RT	120 days	6	748	4	0.97
		RT	120 days	3	732	2	0.95
		WD	72 days	6	651	6	0.84
	Solution 2	60 °C	70 days	6	493	4	0.64
		40 °C	70 days	4	688	10	0.89
		RT	120 days	4	664	7	0.86
		WD	72 days	6	561	8	0.73
	Solution 3	60 °C	70 days	5	564	6	0.73
		40 °C	70 days	4	711	8	0.92
		RT	120 days	4	713	10	0.92
		WD	72 days	5	618	11	0.80
	Solution 4	60 °C	70 days	6	572	2	0.74
		40 °C	70 days	6	754	4	0.98
		RT	120 days	4	751	4	0.97
		WD	72 days	5	666	5	0.86
	Solution 5	FT	300 cycles	4	798	6	1.04
GFRP2 bars	Unconditioned			5	925	3	1.00
	Solution 2	60 °C	60 days	3	482	0	0.52
		40 °C	60 days	4	638	4	0.69
		RT	90 days	4	591	14	0.64
		RT	120 days	4	507	7	0.55
	Embedded in NC and then immersed in Solution 1	RT	90 days	3	836	5	0.90
	Embedded in NC and then immersed in Solution 3	60 °C	90 days	3	566	4	0.61
CFRP bars	Unconditioned			5	2588	5	1.00
	Solution 2	60 °C	70 days	4	2476	9	0.96

Notes: RT = room temperature which was about 20 °C; FT = freezing and thawing cycles as described before; WD = nine wetting and drying cycles as described before; COV = coefficients of variation.

in Fig. 8(b). Similar “whitening” was also observed previously by Bank et al. [19] for GFRP bars embedded in concrete. But GFRP bars embedded in HPC did not exhibit such change in the surface as shown in Fig. 8(c). This may be explained by the presence of slag and silica fume and much lower water–cementitious ratio for HPC, resulting in lower alkalinity of pore solutions and permeability than those in NC, and therefore, inducing less degradation for specimens embedded in HPC than NC.

4.1. Tensile test results

The tensile test results of unconditioned and conditioned specimens are summarized in Table 5. Each data represents three to six test results. During tensile test, all three types of specimens (GFRP1, GFRP2, and CFRP bars) showed an elastic behavior up to failure and failed through the rupture of fibers. The failure of both types of GFRP bars was accompanied by the separation of fibers and the rupture of the fiber bundle with spiral pattern on the deformed sur-

face of bars as shown in Fig. 9. The failure of CFRP bars was catastrophic with a sudden release of stored elastic energy. The typical failure mode of CFRP bars is shown in Fig. 10. Similar tensile failure modes of FRP bars were also observed by Micelli and Nanni [9]. For GFRP bars that were embedded in concrete, the tensile failure initiated within the embedded sections.

After 70 days exposure to Solution 2 at 60 °C, CFRP bars experienced only 4% reduction in tensile strength. While significant degradation was observed in conditioned GFRP1 and GFRP2 specimens, especially for those exposed to solutions at 60 °C. About 36% reduction in tensile strength was found for GFRP1 bars after 70 days immersion in Solution 2 at 60 °C. In case of GFRP2 bars, about 50% reduction was observed for specimens after 60 days immersion in Solution 2 at 60 °C. Similarly, the GFRP2 specimens embedded in NC during the immersion also exhibited a tensile strength reduction of about 40% after 90 days exposure to Solution 3 at 60 °C. It is evident that the temperature of 60 °C had significantly accelerated the

Table 6
Environmental conditions and results for short beam test

Bar types	Environmental exposure			No. of specimens	Mean value of ILSS (MPa)	COV (%)	Retention
GFRP1 bars	Unconditioned			6	43	6	1.00
	Solution 1	60 °C	45 days	5	41	7	0.96
		FT	300 cycles	5	44	4	1.03
		FT	600 cycles	5	40	6	0.93
		FT	600 cycles	5	44	3	1.03
	Solution 2	60 °C	45 days	6	39	6	0.92
	Solution 3	60 °C	45 days	6	42	2	0.99
	Solution 5	FT	300 cycles	5	44	3	1.03
		FT	600 cycles	5	42	3	0.97
	Embedded in NC and then immersed in Solution 1	RT	60 days	5	43	7	1.00
	Embedded in HPC and then immersed in Solution 1	RT	60 days	4	42	5	0.97
	Embedded in NC and then immersed in Solution 3	60 °C	60 days	5	38	6	0.88
	Embedded in HPC and then immersed in Solution 3	60 °C	60 days	4	41	7	0.95
GFRP2 bars	Unconditioned			6	43	5	1.00
	Solution 1	60 °C	45 days	6	39	10	0.90
		FT	300 cycles	5	37	5	0.86
		FT	600 cycles	5	39	4	0.90
	Solution 2	60 °C	45 days	6	34	6	0.79
	Solution 3	60 °C	45 days	6	40	6	0.94
	Solution 4	60 °C	45 days	5	39	6	0.90
	Solution 5	FT	300 cycles	5	42	4	0.99
		FT	600 cycles	5	39	4	0.90
CFRP bars	Unconditioned			9	50	5	1.00
	Solution 2	FT	600 cycles	5	48	2	0.96

Notes: FT = freezing and thawing cycles as described before; COV = coefficients of variation.

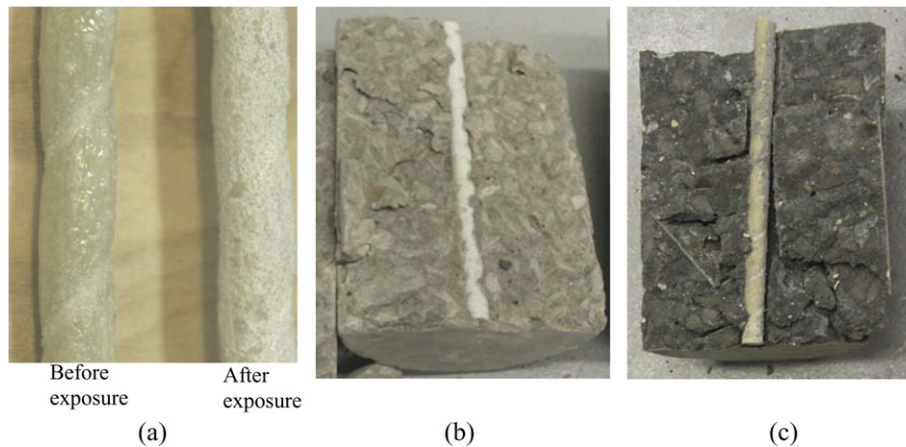


Fig. 8. GFRP bars: (a) GFRP bars before and after exposure to solutions; (b) GFRP bars embedded in NC after splitting; (c) GFRP bars embedded in HPC after splitting.

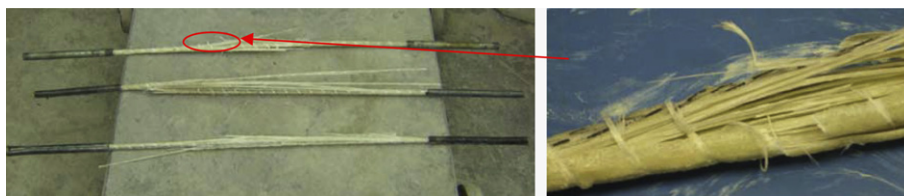


Fig. 9. Typical failure mode of GFRP bars.

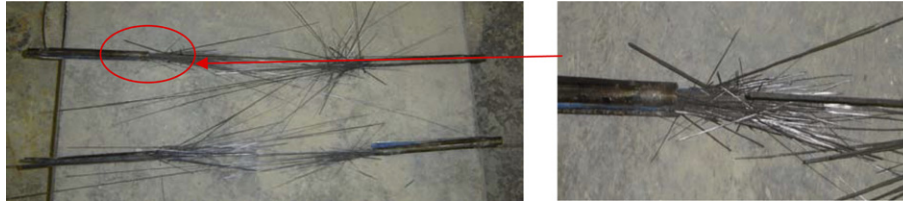


Fig. 10. Typical failure mode of CFRP bars.



Fig. 11. Typical failure mode (shear failure) of GFRP specimens in short beam test.

attack of environments on GFRP bars. But the accelerating effect of the 40 °C temperature was not so obvious.

It is also observed that for all four solutions, the degradation effect of 72-days WD cycles was less significant than 70-days continuous immersion at 60 °C. This may be due to reduced exposure time in solutions at 60 °C for WD cycles.

4.2. Short beam test results

The experimental results for short beam tests are listed in Table 6. Each data is an average of four to six measurements. Most FRP specimens in short beam test failed in shear mode (horizontal cracks along the mid-plane of the specimens) as shown in Fig. 11. Since shorter span to diameter ratio was used by Micelli and Nanni [9], splitting of specimens (failure due to vertical crack) was reported in their study. The typical load vs. deflection relationships for GFRP1, GFRP2 and CFRP specimens are plotted in Figs. 12–14, respectively. Due to the low stiffness and strength of matrix compared to fibers for GFRP bars, there was some deformation of the specimens directly under the loading head at the initial stage of loading. This explains why the curves of load vs. deflection of GFRP bars have lower slope at the initial stage in Figs. 12 and 13, but after

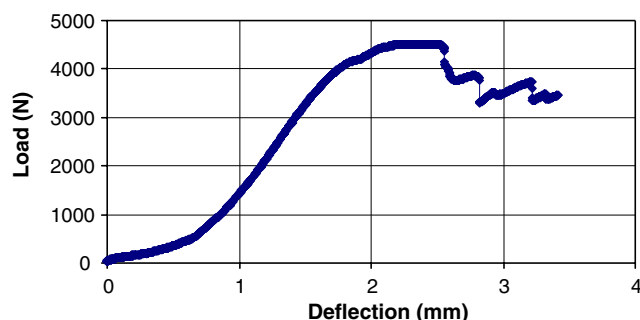


Fig. 12. Typical load vs. deflection of GFRP1 bars in short beam test.

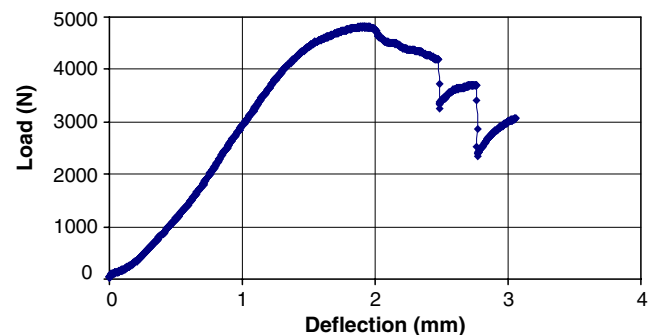


Fig. 13. Typical load vs. deflection of GFRP2 bars in short beam test.

the initial deformation, there was an increase in stiffness as shown in the figures. The stiffness eventually decreased because horizontal cracks (shear failure) started from the mid-plane under the loading head. Once the cracks reached the end of the specimen, shear mode failure occurred. After the shear failure (horizontal cracks), the specimen could still carry some load due to flexural strength of the specimens. Figs. 12–14 further indicate that GFRP specimens had more toughness (area under load–deflection curve) than CFRP specimens.

Since GFRP1 and GFRP2 bars had the same matrix, the initial ILSS for both types of bars were the same. After 45 days exposure to Solution 2 at 60 °C, GFRP1 specimens had a reduction of 8% in ILSS, while a reduction of about 20% was observed for GFRP2 bars. Almost no reduction was found for GFRP1 bars embedded in NC after 60 days exposure to Solution 1 at room temperature, while a reduction of 12% was found for those exposed to Solution 3 at 60 °C. But almost no degradation was observed for all GFRP1 bars embedded in HPC as anticipated from earlier visual examination.

It was also found that generally FT cycles had little effect in accelerating the attack of Solutions 5 and 1 on FRP bars. Surprisingly, after exposure to 300 cycles of FT, some FRP bars even gained some strength, which

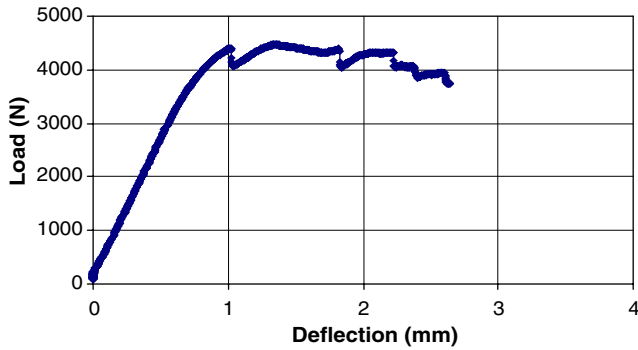


Fig. 14. Typical load vs. deflection of CFRP bars in short beam test.

was also observed previously by Stone et al. [23]. It is possible that some initial range of FT cycles may actually reduce residual stresses. But the degradation effects of FT cycles increased with the cycle numbers. It was also apparent that when combined with FT cycles, the adverse effect of Solution 5 (alkaline solution with Cl^-) was less than that of Solution 1 (tap water).

4.3. Pullout test results

The specimens after pullout test are shown in Fig. 15. All specimens with HPC failed due to the splitting of the concrete cylinders prior to the bars being pulled out. This was because of the very high force needed to pull the bars out of the concrete. So this set of data could not be used to determine the pullout strength of bars. The typical load vs. displacement (recorded by LVDT as shown in Fig. 7) curves are shown in Figs. 16–19. As for HPC, the specimens failed by splitting before the slipping of the bars occurred, for which the load dropped to zero immediately after concrete splitting as shown in Figs. 16 and 17. For embedment within NC, as shown in Figs. 18 and 19, all the specimens failed due to bond failure. For this type of concrete, after the slip, the specimens could still carry some load due to the friction between the helically wrapped and sand coated surface of bars and the concrete. This phenomenon was more pronounced in the case of larger diameter bar (No. 4) as shown in Fig. 19.



Fig. 15. Specimens after the pullout test.

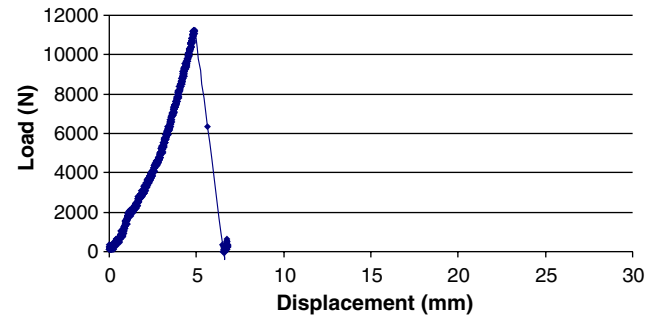


Fig. 16. The load vs. displacement curve of specimen with HPC and GFRP1 #3 bar after 60 days exposure in curing tank.

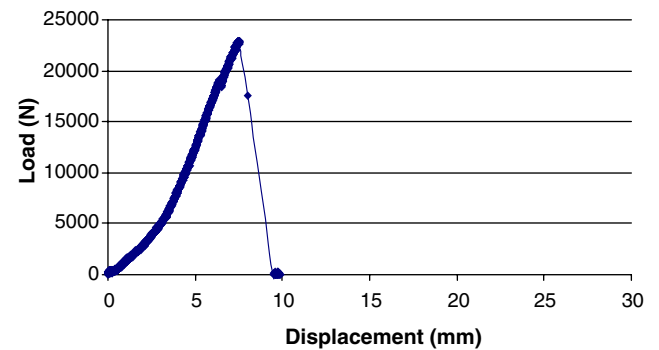


Fig. 17. The load vs. displacement curve of specimen with HPC and GFRP1 #4 bar after 60 days exposure in Solution 3 at 60 °C.

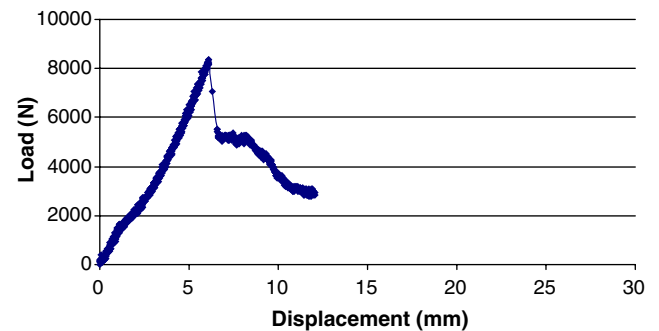


Fig. 18. The load vs. displacement curve of specimen with NC and GFRP1 #3 bar after 60 days exposure in curing tank.

The degradation effects of alkaline solution at 60 °C on bond strength are shown in Fig. 20 (each column is based on an average of two test results). It is observed that the elevated temperature and alkalinity accelerated the degradation of bond strength. The specimens retained about 88% of bond strength of the companion specimens after the 60 days exposure. The same amount of retention (see Table 6) was observed for ILSS of GFRP1 bars embedded in NC after the same exposure (the specimens for short beam tests and pullout tests came from the same concrete cylinder as shown in Fig. 6). This indicates that there is a possible correlation between material degradation of FRP bars and decrease in bond strength, which was also

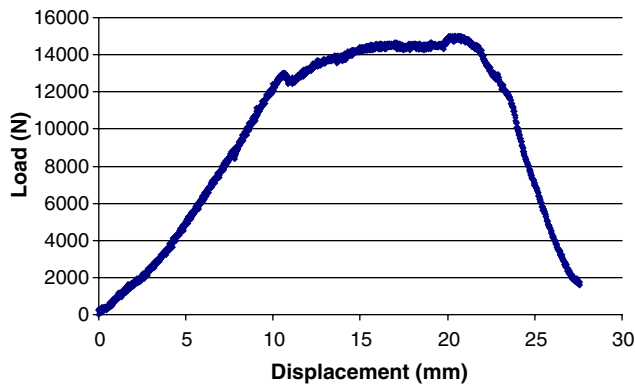


Fig. 19. The load vs. displacement curve of specimen with NC and GFRP1 #4 bar after 60 days exposure in curing tank.

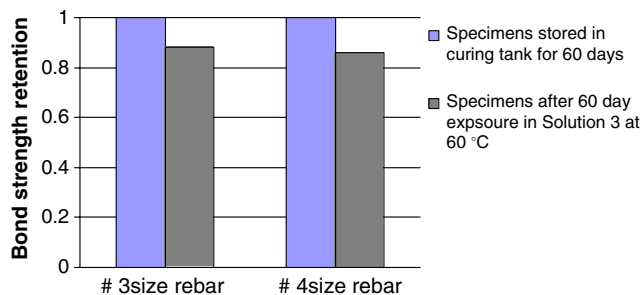


Fig. 20. Bond strength retention of GFRP1 bars in NC.

reported by Bank et al. [19] based on the decrease in bond strength and corresponding visual inspection and scanning electron micrographs of FRP bars extracted from concrete in their study.

4.4. Summary

Though initially GFRP2 bar had higher tensile strength and same ILSS compared to GFRP1 bar, it underwent more loss in strength due to environmental attack. Since these two types of bars had same matrix and were made following similar manufacturing processes, the inferior durability performance of GFRP2 bar may be attributed to lower durability performance of glass fibers and/or fiber/matrix interphase. Also, it was found that the ILSS of GFRP bars deteriorated at a slower rate than tensile strength.

The Solution 2 was the most aggressive environment to cause the deterioration of FRP bars. It was found that generally the aggressiveness of the solutions decreased in the following order: Solution 2 being the most severe followed by Solutions 3, 1, and 4. However, at 60 °C, the Solution 1 was slightly more aggressive than Solution 3. This phenomenon may possibly be explained as follows. At the temperature of 60 °C, more water molecules in Solution 1 might have diffused into FRP bars, leading to more degradation, than simulated HPC pore Solution 3. However, further research is needed to support this argument. It was also ob-

served that Solution 4 was slightly less aggressive than Solution 1. This may be explained by the characteristics of the selected GFRP specimens known to be impervious to chloride ions. Moreover, it is widely accepted that distilled water is more aggressive to composites than seawater [16].

From the results of GFRP bars embedded in concrete, it was found that the attack of concrete environment may also be accelerated by immersing FRP-concrete specimens in solutions at elevated temperature. Little degradation was observed for specimens embedded in HPC, which provided a less aggressive environment than NC.

A comparison of displacements measured on bars during pullout test (as shown in Figs. 16–19) indicated that for a given amount of tensile force, the FRP bars within NC displaced more than those embedded in HPC, although in this case failure was caused by concrete fracture. The significantly higher load of resistance against bond failure, and lower displacements for a given level of load, indicate that HPC by virtue of its stronger matrix offered significantly higher resistance to displacement. This implies that the bond strength (or pullout strength) of FRP bars is dependent not only on the shape, surface characteristics and size of FRP bars, but also on the type of concrete in which it is embedded.

5. Conclusions and recommendations

This study is part of an ongoing durability research program on fiber-reinforced polymer (FRP) reinforcing bars for concrete structures. From the test results presented in this paper, the following concluding remarks and recommendations can be drawn.

Durability performance of GFRP bars could be improved by selecting the appropriate constituents of the composites. Alkali resistant fibers with proper sizing, if available, may be helpful in further enhancing the durability performance of GFRP bars in alkaline environments. In addition to the constituents of FRP bars in general, the quality of manufacturing and particularly fiber/matrix interphase does influence the durability of FRP bars, as indirectly observed in this study.

Elevated temperature accelerated the degradation of both bare FRP and FRP-concrete specimens. The alkalinity of simulated NC environment (with a pH value of 13.6) was the most aggressive condition for bare FRP bars, and more detrimental than for bars embedded in concrete. HPC environment was less aggressive to GFRP bars than NC environments. Both WD and FT cycles had negligible effects on degradation process, and therefore, these two exposure conditions are not recommended as critical for performance tests.

Since the bond strength between FRP bars and concrete was found to degrade after environmental exposures, the durability performance of bond strength should also be investigated extensively. A possible correlation was found between degradation of ILSS of FRP bars and decrease

in bond strength between FRP bars and concrete. Additionally, the bond failure of FRP bar embedded in concrete was strongly influenced by the types of concrete. Therefore, more representative evaluations of performance of FRP reinforced concrete structures can be made through the testing of FRP embedded in concrete of similar constituents to be used in real-life construction. The bond test method (used in this study) of FRP-concrete for HPC should be modified in order to achieve complete pullout before splitting of the concrete due to its high bond capacity. A larger size concrete specimen may help to eliminate the splitting before pullout. Further study on FRP-concrete for bond tests and flexure tests should be conducted before final conclusions can be made on durability performance.

Since the bare FRP specimens were directly exposed to the solutions and FRP-concrete specimens were kept in saturated conditions until testing, the reported experimental results should be considered as conservative. It is recommended that FRP-concrete combinations be considered for testing in addition to bare FRP bars to evaluate the durability of FRP reinforcement, since the protocols of bare FRP testing are too harsh for realistic conditions. Long-term data including those exposed to outdoor environments are greatly valuable in order to validate short-term accelerated aging test results. Moreover, coupling effects of sustained load and environmental exposures should also be included in future studies in order to simulate real-life conditions.

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